higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

30 July 2024 (X-paper)
09:00–12:00

Scientific calculators and drawing instruments may be used.

This question paper consists of 6 pages and a formula sheet of 5 pages.

072Q1G2430

Copyright reserved

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show all intermediate steps and simplify where possible.
5. All final answers must be rounded off to THREE decimal places.
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Sketches must be large, neat and fully labelled
8. Use only a black or blue pen.
9. Write neatly and legibly.

Copyright reserved

QUESTION 1

1.1 Given:

$$\ln y = \lim_{x \to 0} \frac{\tan x - \sin x}{3x^2}$$

Calculate the numerical value of:

1.1.1 $\ln y$ (3)

1.1.2 y (1)

1.2 Determine the value(s) of x for which $f(x)$ is discontinuous if:

$$f(x) = \frac{e^x - 1}{\sin x}, \quad [0°; 270°]$$ (2)

[6]

QUESTION 2

2.1 Determine $\frac{dy}{dx}$ if $f(x) = \cos x$ using first principles.

Show all intermediate steps

HINT: $\lim_{x \to 0} \frac{\cos h - 1}{h} = 0$ and $\lim_{x \to 0} \frac{\sin h}{h} = 1$ (4)

2.2 Determine $\frac{dy}{dx}$ in each of the following cases:

(Simplification NOT required)

2.2.1 $y = \sin^6(\ln(4x^5))$ (3)

2.2.2 $y = \left(\frac{x^2 + 4}{x^3 - 3x + 6}\right)^4 + \sqrt{2x - 5}$ (5)

2.3 Calculate $\frac{dy}{dx}$ if $y = (\cos x)^{-2x}$ with aid of logarithmic differentiation (5)

2.4 Determine $\frac{dy}{dx}$ of implicit function:

$$\tan 3y - y^3 \cos x + 3xy^2 = -4$$ (5)

[22]

Copyright reserved

QUESTION 3

3.1 Given:

$$f(x) = x^3 - 3x^2 - 3x + 6$$

3.1.1 Determine the coordinates of the turning points of $f(x)$. (3)

3.1.2 Draw up a table of x and $f(x)$, where x ranges from $x = -2$ to $x = 4$. (2)

3.1.3 Draw a neat graph of $f(x)$ between these values and show the turning points on it. (2)

3.1.4 Use the table and the graph to estimate a value for the best root between $x = -2$ and $x = -1$ of the equation $x^3 - 3x^2 - 3x + 6 = 0$. Then use Taylor's /Newton's method ONCE to determine a better approximation of this root. (Root correct to THREE decimal figures.) (3)

3.2 The volume of a tree is given by $V = \frac{1}{12\pi}C^2h$, where C is the circumference of the tree at ground level (in meters) and h is the height of the tree (in meters).

Both C and h are functions of time (in years).

3.2.1 Find a formula for $\frac{dV}{dt}$. What does it represent? (3)

3.2.2 Suppose that the circumference grows at a rate of 0,2 meters per year and the height grows at a rate of 4 meters per year. How fast does the tree grow when the circumference is 5 meters, and the height is 22 meters? (3)

3.3 Find the positive number where the sum of that number and its reciprocal is minimum. (4)

[20]

QUESTION 4

4.1 Determine $\int y\,dx$ in each of the following cases:

4.1.1 $y = \cos 3x \sin(-7x)$ (3)

4.1.2 $y = \sin^4 x \operatorname{cosec} x$ (5)

4.1.3 $y = \dfrac{x^3 - 3x^2 - 10x + 24}{x + 4}$ (5)

4.1.4 $y = x^3 \ln 4x$ (4)

Copyright reserved

4.2 Determine $\int y\,dx$ by resolving the integral into partial fractions:

$$y = \frac{2x-1}{(x+3)^3}$$ (7)

[24]

QUESTION 5

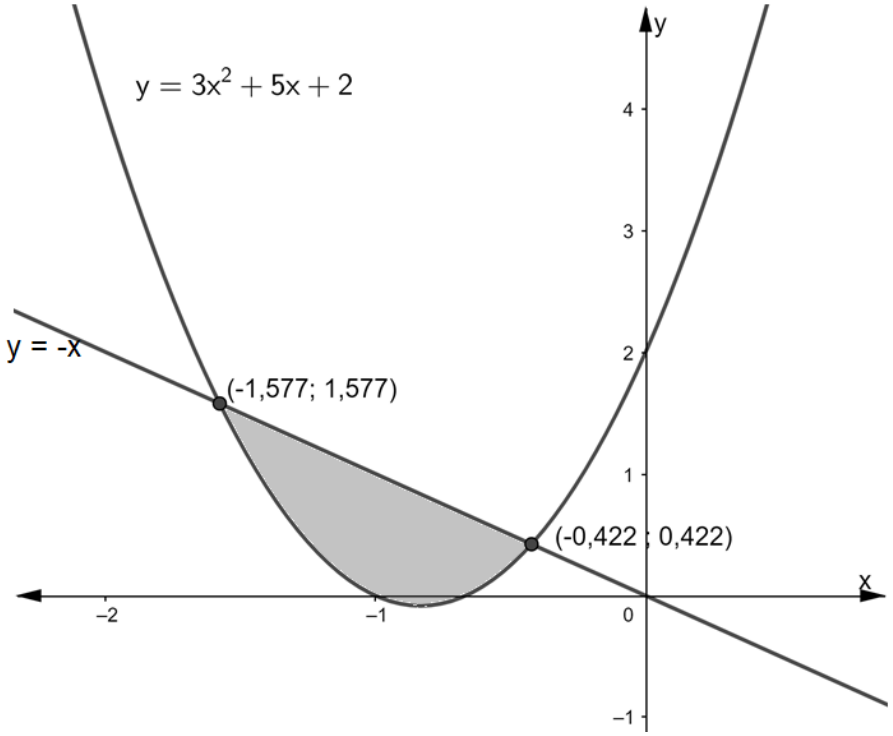
5.1 If $f(2) = 5$ and $f(0) = 1$, then evaluate $\int_0^2 f'(x)\,[f(x)]^2\,dx$ (4)

5.2 Show that if $f(t) = p$, then the integral:

$$\int_0^{\infty} e^{-2t} f(t)\,dt = \frac{p}{2}$$

If $f(2) = 5$ and $f(0) = 1$, then evaluate $\int_0^2 f'(x)\,[f(x)]^2\,dx$ (3)

5.3 Given below is the graphs of the functions: $y = 3x^2 + 5x + 2$ and $y = -x$, showing the points of intersection and the area bounded between the curves.



5.3.1 Calculate the magnitude of the bounded area. (3)

5.3.2 Calculate the volume of the solid of revolution formed when the area in QUESTION 5.3.1 is rotated about the x-axis. (5)

5.4 Calculate the moment of inertia of a circular lamina of radius $\frac{r}{2}$ and mass m about an axis through its centre and perpendicular to the plane of the lamina. (5)

[20]

QUESTION 6

6.1 Determine the particular solution of:

$$\frac{dy}{dx} = \frac{2xy}{y^2 - 1} \quad y(2) = 1$$ (5)

6.2 Determine the general solution of:

$$14 + 2x\frac{d^2y}{dx^2} = 2x^3 - 6x^2$$ (3)

[8]

TOTAL: 100

Copyright reserved

MATHEMATICS N5

FORMULA SHEET

Any applicable formula may also be used.

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \text{cosec}^2 x$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$\sin^2 A = ½ - ½ \cos 2A$

$\cos^2 A = ½ + ½ \cos 2A$

$\sin (A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$\sin A \cos B = ½ [\sin (A + B) + \sin (A - B)]$

$\cos A \sin B = ½ [\sin (A + B) - \sin (A - B)]$

$\cos A \cos B = ½ [\cos (A + B) + \cos (A - B)]$

$\sin A \sin B = ½ [\cos (A - B) - \cos (A + B)]$

$$\tan x = \frac{\sin x}{\cos x} ; \sin x = \frac{1}{\text{cosec } x} ; \cos x = \frac{1}{\sec x}$$

Copyright reserved

BINOMIAL THEOREM

$$(x + h)^n = x^n + nx^{n-1}\, h + \frac{n(n-1)}{2!} x^{n-2}\, h^2 + \ldots$$

DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x) \,.\, v(x)$$

$$\frac{dy}{dx} = u \,.\, \frac{dv}{dx} + v \,.\, \frac{du}{dx}$$

$$= u \cdot v' + v \cdot u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v \,.\, \frac{du}{dx} - u \,.\, \frac{dv}{dx}}{v^2}$$

$$= \frac{v \cdot u' - u \cdot v'}{v^2}$$

CHAIN RULE

$$y = f(u(x))$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Copyright reserved

$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1} + c$
a	0	$ax + c$
e^x	e^x	$e^x + c$
a^x	$a^x.\ln a$	$\frac{a^x}{\ln a} + c$
$\log_e x$	$\frac{1}{x}$	—
$\log_a x$	$\frac{1}{x \ln a}$	—
$\sin x$	$\cos x$	$-\cos x + c$
$\cos x$	$-\sin x$	$\sin x + c$
$\tan x$	$\sec^2 x$	$\ln(\sec x) + c$
$\cot x$	$-\text{cosec}^2 x$	$\ln(\sin x) + c$
$\sec x$	$\sec x \cdot \tan x$	$\ln[\sec x + \tan x] + c$
$\text{cosec } x$	$-\text{cosec } x \cdot \cot x$	$\ln(\text{cosec } x - \cot x) + c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	—
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	—
$\tan^{-1} x$	$\frac{1}{1+x^2}$	—
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	—
$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$	—
$\text{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$	—

Copyright reserved

$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
$\frac{1}{\sqrt{a^2 - x^2}}$	—	$\sin^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{a^2 + x^2}$	—	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{x\sqrt{x^2 - a^2}}$	—	$\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$
$\sqrt{a^2 - x^2}$	—	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{2}\sqrt{a^2 - x^2} + c$
$\frac{1}{x^2 - a^2}$	—	$\frac{1}{2a} ln\left(\frac{x - a}{x + a}\right) + c$
$\frac{1}{a^2 - x^2}$	—	$\frac{1}{2a} ln\left(\frac{a + x}{a - x}\right) + c$

INTEGRATION

$\int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int f'(x)\, g(x)\, dx$

$$\int \frac{f'(x)}{f(x)}\, dx = \ln f(x) + c$$

$$\int [f(x)]^n f'(x)\, dx = \frac{[f(x)]^{n+1}}{n+1} + c$$

$$\frac{f(x)}{(ax+b)(cx+d)} = \frac{A}{ax+b} + \frac{B}{cx+d}$$

$$\frac{f(x)}{(x+a)^n} = \frac{A}{(x+a)} + \frac{B}{(x+a)^2} + \frac{C}{(x+a)^3} + \cdots \frac{Z}{(x+a)^n}$$

APPLICATIONS OF INTEGRATION

AREA

$$A_x = \int_a^b y\,dx \; ; \; A_x = \int_a^b (y_1 - y_2)\,dx$$

$$A_y = \int_a^b x\,dy \; ; \; A_y = \int_a^b (x_1 - x_2)\,dy$$

Copyright reserved

VOLUME

$$V_x = \pi \int_a^b y^2 dx \; ; V_x = \pi \int_a^b \left(y_1^2 - y_2^2\right) dx$$

$$V_y = \pi \int_a^b x^2 dy \; ; V_y = \pi \int_a^b \left(x_1^2 - x_2^2\right) dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2 dA \; ; I_y = \int_a^b r^2 dA$$

MOMENTS OF INERTIA

Mass = density × volume

$M = \rho V$

DEFINITION: $I = m r^2$

GENERAL: $I = \int_a^b r^2 dm = \rho \int_a^b r^2 dV$

Copyright reserved